

FLIGHT SIMULATOR X: ACCELERATION KEY COMMAND CHANGES



Games
for Windows®

To add *Flight Simulator X: Acceleration's* new commands to your key mappings, click Yes during Setup. If you click No and later change your mind, you can revise key assignments in the Controls dialog box (on the **Options** menu, point to **Controls**, and then choose **Assignments**).

Helicopter Hoist/Sling Operation

Function	Key Command	Previous Function
Hoist Arm (Extend / Retract)	CTRL + U	None
Hoist Cable (Lower)	J	Jet Starter
Hoist Cable (Raise)	K	Point of Interest
Hoist / Sling Hook (Attach / Release)	O	Strobe Lights On

Aircraft Carrier Operations

Function	Key Command	Previous Function
Tail Hook (Extend / Retract)	SHIFT + Q	None
Takeoff Assist (Arm / Disarm)	SHIFT + I	None
Launch Bar (Extend / Retract)	SHIFT + U	None
Takeoff Assist (Trigger)	SHIFT + Spacebar	None

Extra 300 Operation

Function	Key Command	Previous Function
Nitrous Oxide Injection (On / Off)	SHIFT + C	None

Microsoft
game studios

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Important Health Warning About Playing Video Games

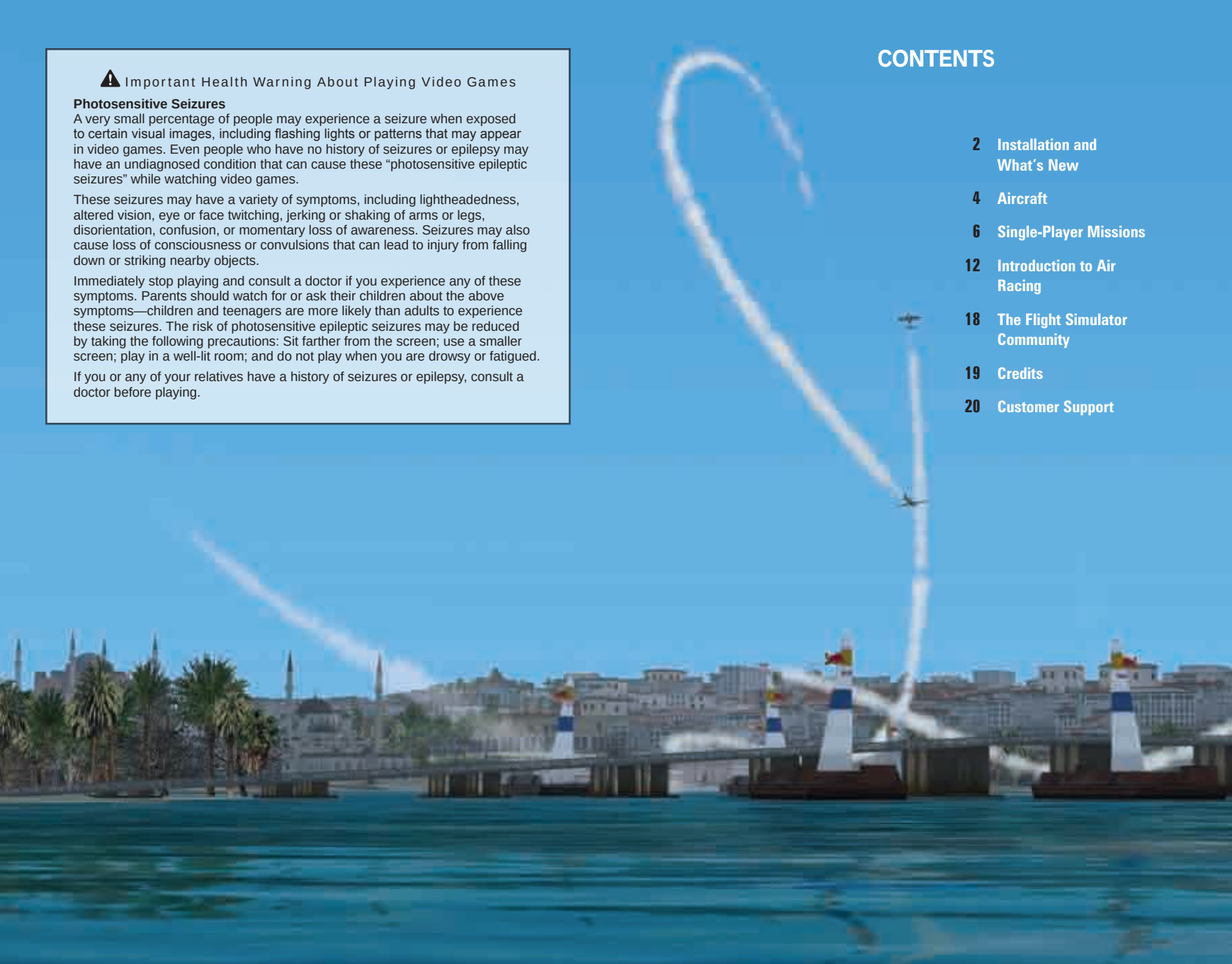
Photosensitive Seizures

A very small percentage of people may experience a seizure when exposed to certain visual images, including flashing lights or patterns that may appear in video games. Even people who have no history of seizures or epilepsy may have an undiagnosed condition that can cause these "photosensitive epileptic seizures" while watching video games.

These seizures may have a variety of symptoms, including lightheadedness, altered vision, eye or face twitching, jerking or shaking of arms or legs, disorientation, confusion, or momentary loss of awareness. Seizures may also cause loss of consciousness or convulsions that can lead to injury from falling down or striking nearby objects.

Immediately stop playing and consult a doctor if you experience any of these symptoms. Parents should watch for or ask their children about the above symptoms—children and teenagers are more likely than adults to experience these seizures. The risk of photosensitive epileptic seizures may be reduced by taking the following precautions: Sit farther from the screen; use a smaller screen; play in a well-lit room; and do not play when you are drowsy or fatigued.

If you or any of your relatives have a history of seizures or epilepsy, consult a doctor before playing.

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INSTALLATION

You must have Microsoft® *Flight Simulator X* installed before installing *Flight Simulator X: Acceleration*. Insert the disc into your computer's disc drive, and then follow the on-screen instructions. If Setup does not start automatically, follow these steps.

To install on Microsoft Windows® XP

1. On the **Start** menu, click **Control Panel**.
2. In the **Category View**, under **Pick a Category**, click **Add or Remove Programs**.
3. Under **Pick a Task**, click **Add a Program**.
4. In the **Add or Remove a Program** dialog box, click the **CD or Floppy** button, and then follow the on-screen instructions to install the game.

To install on Microsoft Windows Vista®

1. On the **Start** menu, click **Computer**.
2. Under **Devices with Removable Storage**, double-click the disc drive icon.
3. In the **User Account Control** dialog box, click **Allow**, and then follow the on-screen instructions to install the game.

WHAT'S NEW?

Flight Simulator X: Acceleration includes three new aircraft, more than 20 challenging single-player missions, and 19 competitive multiplayer missions. These new experiences include jet carrier operations, fixed-wing and helicopter transport missions, search and rescue (SAR) missions, and single and multiplayer racing.

Windows Vista Features

- ▶ **Games Explorer** The Games Explorer (on the **Start** menu) lists all games installed on your PC. Icons offer easy access plus information for sorting and organizing your games. Right-click an icon for a menu of play tasks and online options, such as linking to the game's Web site. Use the Games Explorer command bar to access the player community for each game; get help, support, and important publisher updates; and easily add or remove games.
- ▶ **Parental Controls** Windows Vista parental controls allow parents to decide which games their children can play, which programs they can use, and which Web sites they can visit — and when. You can also review activity reports that show how your children have been using the computer.
- ▶ **DirectX 10 Support** *Flight Simulator X: Acceleration* uses the power of Windows Vista and DirectX®10 to deliver breathtaking graphics and dynamic environments to immerse you in the world of flight.

New Multiplayer Functionality

The new multiplayer experience in *Flight Simulator X: Acceleration* focuses on racing. You can limit participants to include only your friends or take on all comers. During a race, you'll see a map showing the race position of each participant. The system tracks who is winning and displays an end-of-race dialog box showing all finishers and their times for the race.

AIRCRAFT

In addition to the existing stable of *FSX* aircraft, *Flight Simulator X: Acceleration* puts you in the pilot's seat of three exciting new aircraft.

- ▶ **Strap yourself into a race-modified P-51D Mustang and scorch the course at Reno.** With clipped wingtips, reduced weight, and more than twice the horsepower of the original, this pumped-up P-51 wrings extra performance out of one of the best fighters of World War II. Special racing systems, including ADI (anti-detonation injection of water and alcohol), allow sustained operation at high boost pressures, but you must manage these systems — or blow your engine and lose the race.
- ▶ **Grab some stick time in the Boeing F/A-18A Hornet fighter/attack jet.** This fast, maneuverable twin-engine single-seater is a versatile front-line fighter in many air forces, and it also powers the U.S. Navy Blue Angels demonstration team through some of the world's most challenging aerobatic routines. With its two powerful jet engines and six stages of afterburner, the Hornet has performance to spare.
- ▶ **Master the powerful, rugged AgustaWestland EH101 medium-lift helicopter** and its hoist and slingload systems to deliver cargo or rescue shipwreck survivors. Its three turboshaft engines, rugged construction, and redundant systems give it the power to lift heavy loads and the reliability to perform in harsh conditions. The rest is up to you!



P-51D Mustang



Boeing F/A-18A Hornet



AgustaWestland EH101



SINGLE-PLAYER MISSIONS

New single-player missions put you in the cockpit for some exciting new experiences. Fly F/A-18 carrier operations, transport people and cargo in some tricky locations and situations, master the intricacies of helicopter hoist and slingload operations, and find and rescue people in danger. You can also race a variety of aircraft in some of the world's most demanding venues. Victory means getting the most out of your aircraft — and yourself.

Carrier Operations

Flying an F/A-18 Hornet off of — and back onto — an aircraft carrier is one of the most challenging experiences in *Flight Simulator X: Acceleration*, and a chance to master some very specialized skills. You'll launch off the deck from a catapult; learn to fly the visual traffic pattern; master the complexities of throttles, landing gear, arrestor hook, flaps, and speed brake; and then snag the arrestor cables, ending the adventure as abruptly as it began. If you don't get it right, you'll get a wave-off to go around the pattern and try it again.



TIPS & TRICKS

- ▶ When positioned over a catapult, press **SHIFT+U** to extend the Launch Bar, and then press **SHIFT+I** to attach to the catapult.
- ▶ When you're ready to take off, advance the throttles to full military power (without afterburner), and then press **SHIFT+SPACEBAR** to fire the catapult.
- ▶ The afterburner uses a lot of fuel; watch your fuel level when flying long sorties.
- ▶ If you can't see the top of the deck throughout your approach, you're too low.
- ▶ Press **SHIFT+Q** to extend/retract the tailhook. As soon as you hit the carrier deck, add full thrust for a go-around. If you catch a cable, reduce thrust to idle as soon as the aircraft slows to a stop.
- ▶ See the Carrier Tutorial mission briefing for more information on carrier operations; see F/A-18A Flight Notes for more information on flying the Hornet.



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Transporting People and Cargo

Transporting people and cargo sounds routine, but these missions offer some unique challenges. Whether you're flying a Beech Baron, a Cessna Caravan, or an ancient DC-3, a "routine flight" can quickly become anything but routine. This is even more true when the mission involves flying a helicopter and using its hoist and slingload systems to deliver people or cargo.

The EH101 medium-lift helicopter is a powerful and versatile workhorse, but, like any helicopter, it can be difficult to fly. Picking up and setting down heavy loads at a precise spot on the ground or water sounds easy — until you try it. Tasks such as snagging a rocket payload from the ocean, delivering an antenna array to New Mexico's Sandia Crest, or getting a replacement cable car pulley to the top of Sugarloaf Mountain in Rio push your piloting skills to the limit. These missions test your patience and precision as you compensate for gusting winds and the weight of the load swinging beneath you. Sometimes releasing the load is your only option, but your boss may not like it!



TIPS & TRICKS

- ▶ When you are ready to hook onto a load, press **CTRL+U** to extend the hoist arm, and then press **J** to lower the cable. When the cable is close to the object, the hook automatically engages. Press **K** to raise the cable. Press **O** to release the load and deactivate cable hook-attach mode. Pressing **O** again prepares the hook to engage another load.
- ▶ When transporting an object on a cable, be sure to fly high enough to avoid hitting anything and destroying the load.
- ▶ As you fly, the slingload acts like a pendulum, pulling on the helicopter as it swings. The heavier the load, the more the pendulum effect is exaggerated. The more gently and smoothly you fly, the less motion you impart to the load.
- ▶ For more Tips and Tricks, see the Slingload Tutorial and the next section, Search and Rescue (SAR) Missions.



Search and Rescue (SAR) Missions

On search and rescue missions, the stakes are high and time is limited. Hovering 100 feet above a sinking vessel in the stormy North Sea and using the helicopter hoist to rescue the crew calls for a cool head and a steady hand. Wind gusts and waves add to the difficulties. Or try plucking an accident victim off a bridge on Florida's Overseas Highway and transporting the person to the nearest hospital. A geology crew stranded in the crater of Mount St. Helens may depend on your skill to get them out before the mountain erupts again. Finding a pilot downed in a snowy valley or a kayaker lost in the waters off Florida requires keen eyes and top-notch flying skills to get them out of sticky situations. These missions offer some edgy piloting experiences with unique risks and rewards.



TIPS & TRICKS

- ▶ Using a hoist cable is similar to using a slingload, except that the load is hanging off the side of the helicopter and thus tends to pull it in that direction.
- ▶ Most of the time you lower a rescue crew member from the hoist, and then pull the crew and victim up into the helicopter. The hoist handles up to 600 pounds, and the cable extends to 300 feet.
- ▶ When a hoist load is at the very top of its travel, press **O** to pull it into the helicopter. If you have a load in the helicopter and the hoist is at the top without a load attached, pressing **O** places the load back onto the cable.
- ▶ Use external views or open a new view in combination with an external view to monitor hoist operations. To learn more, see *Using Views and Windows* in the Learning Center.



INTRODUCTION TO AIR RACING

Pilots deal with all kinds of pressure while flying. Even when the weather is clear and calm, and your plane is in top condition, flying requires concentration. You scan the sky and your panel continually, and manage the various aircraft systems, all while taking in the phenomenal view from the cockpit. Air racing adds new pressures to the mix and calls for an unparalleled level of confidence and control.

The big difference, of course, is time. Racing isn't about getting from here to there on schedule — it's about getting there first. But it takes more than a heavy hand on the throttle to be a successful racer. It requires intelligence and finesse. The tendency is to focus on maximum throttle and aggressive maneuvers, but, despite competitive pressures, the fastest way around any course is also the smoothest. A light hand on the stick and minimal control inputs save time. Flying reactively — responding abruptly to sudden moves by the competition, overcontrolling the aircraft, and then overcorrecting — can put you at the back of the pack.

It takes discipline to fly strategically — and it takes a lot of practice. It all begins with memorizing and mastering the race course. If you're constantly rediscovering what's around the corner, you can't focus on flying smooth laps. Start slowly, and then build up your speed on each successive lap. Soon you'll know every part of the course and will be able to fly every lap at or near the limit of your ability. A consistent, seamless approach can make the difference between being a podium finisher and an also-ran.



TIPS & TRICKS

- ▶ To learn a course and master your aircraft, practice racing single-player missions until you're ready to face live challengers in multiplayer races.
- ▶ Practice lapping the course until you know it intimately. You can also learn the course by following the race leader for a few laps before making your move.
- ▶ Make sure the way you handle each part of the course feeds seamlessly into handling the next; fly smoothly and consistently.
- ▶ Make the smallest control inputs required to get around the course quickly.
- ▶ In general, avoid high-G maneuvers; minimal control inputs and smooth, consistent flying give you a better chance of a higher ranking.
- ▶ Especially in Reno air racing, monitor your ADI, cylinder head temperature, manifold pressure, and overall engine health carefully. Run too hot or pile on too much boost, and you won't finish the race.
- ▶ Minimize altitude gain on the course, as climbing slows you down.



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Red Bull Air Race World Series

The Red Bull Air Race World Series is a unique test for aerobatic pilots. You navigate the tight turns and high-G maneuvers of a short, intense aerial slalom course in the fastest possible time while incurring as few penalties as possible. *Flight Simulator X: Acceleration* adds several demanding new Red Bull Air Race-style courses at the same venues as some of the official World Series events. Each course features numbered sets of 20-meter-tall, air-filled pylons, called “air gates,” marked with colored bands. You must pass between blue gates flying straight and level, and between red gates in knife-edge (vertical) flight. You can pass between checkered or white gates at any bank angle. At intervals during the timed flight, you must correctly perform required aerobatic maneuvers. Darting between pylons and performing aerobatics requires speed, skill, smoothness — and total familiarity with the course.

Time penalties can be costly. Lose three seconds for passing gates too high, failing to fly horizontally or vertically through the appropriate gates, and failing to perform a turning maneuver correctly. Lose 10 seconds for touching a pylon with your plane.



TIPS & TRICKS

- ▶ Arrows (the Maneuvers icon) guide you through the more complicated maneuvers. Use the Race Map and Pointer to find your way around the course. For a more challenging experience, press **U** to turn off the Race Map and Pointer.
- ▶ You must memorize these demanding courses before getting down to serious racing. At more than 200 mph, gates come up fast, and there's no extra room for required maneuvers. If you have to think about your next move, you'll blow it!
- ▶ Knife-edge maneuvers are easier to perform while turning.
- ▶ To learn more about the skills you'll use in this mission, see Learning Center articles *Extra 300S Aircraft Information*, *Aerobatics*, and *Competition Air Racing*.



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Reno Air Races

Flying an Unlimited Class race at Reno means high speed and intense concentration. The pear-shaped course is marked by 50-foot-tall telephone poles. "Fly low, go fast, turn left" sounds like a simple formula, but everything happens very quickly, and it's not easy to keep your control inputs smooth and your engine healthy.

Unlimited Class racers are mostly modified WWII fighters, including P-51 Mustangs, F-8F Bearcats, and Hawker Sea Furies. These are the fastest piston-powered, propeller-driven planes in the world; the new P-51 racer produces as much power as a dozen Red Bull racers. Eight of them jockeying for position 60 feet off the ground is an awesome sight — or would be, if you had time to look at anything except the course unreeling at 500 mph and the instruments that indicate if your engine is likely to go the distance.

There's a lot of brute force at play, but it takes finesse to win. You can run an enormous amount of boost in an Unlimited Class racer (up to 150 inches of manifold pressure), but extra performance never comes free. The more throttle and boost you pile on, the hotter the engine runs. Monitor engine parameters, manifold pressure, and overall engine health. Using your ADI keeps cylinder head temperature down, but with a limited supply of the mix, you can't run maximum boost all the time. If you're not careful, you'll lose pistons or throw a rod, and lose the race.



TIPS & TRICKS

- ▶ To lap fast, fly smoothly, pulling as few Gs as possible and making small control inputs.
- ▶ Engage ADI when running above stock P-51 boost limits.
- ▶ Consistently running at high boost, even with ADI on, can damage your engine.
- ▶ Unlike the air gates in a Red Bull Air Race, Reno pylons don't give when you hit them!
- ▶ You are disqualified if you cut across the course, cut pylons, or fly too close to the crowd line.
- ▶ See the Reno Air Racing Practice Tutorial for more information.



Microsoft *Flight Simulator X* is the center of an active worldwide community, and its hub is fsinsider.com.

The FSInsider Web Site

When they're not busy flying the virtual skies, Flight Simulator community members read about aircraft and aviation. The FSInsider Web site is the online destination for those who want to discover the latest Flight Simulator activities, share news, and download new aircraft, missions, and scenery. The site is refreshed periodically with new content, community blogs, and revealing behind-the-scenes glimpses that show how *Flight Simulator X* and the expansion pack achieve their immersive realism.

LINKS ON THE FSINSIDER WEB SITE

- ▶ News and Events
- ▶ From the Team
- ▶ Free Flight
- ▶ Missions and Multiplayer
- ▶ Developer's Corner
- ▶ Performance Tips & Tweaks
- ▶ Partnerships
- ▶ Community Links
- ▶ Product Information
- ▶ Downloads
- ▶ Membership

FSInsider.com is a great place to stay in touch with the Flight Simulator community and the incredible creativity of its members.

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 - Excel Data Corporation
 - Lux
 - ManPower
 - Pasona Tech
 - Sakson & Taylor
 - Things to Come
 - Volt
 - Wise HR
- See the in-game credits for a complete list of *Flight Simulator X: Acceleration* team members.

CUSTOMER SUPPORT

In the U.S., visit support.microsoft.com/games.

In Canada, visit support.microsoft.com/canada.

Outside the U.S. and Canada, visit
support.microsoft.com/international.aspx.

On these Web sites, you can:

- ▶ Download the latest updates.
- ▶ Find Microsoft Knowledge Base articles that target specific issues.
- ▶ View “Show Me How” videos for solving common problems.
- ▶ Access other support services.

Microsoft support services are for technical issues only — please do not use them to request game hints, codes, or cheats.

Chat & E-mail Support: To chat immediately with a Microsoft Support Professional, or to submit your issue by e-mail (e-mail responses may be delayed by several hours), visit the location-appropriate Web site listed above.

Phone Support: To speak with a Microsoft Support Professional (long hold times may be experienced), in the U.S. call (800) 537-8324, and in Canada call (800) 936-8479.

TTY Support: To use Microsoft text telephone, in the U.S. call (800) 892-5234, and in Canada call (866) 857-9850.

Conditions: Microsoft’s support services are subject to then-current prices, terms, and conditions, which are subject to change without notice.

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